CMSC 420: Scapegoat Trees

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1 Problems **2**

1 Problems

1. Given the following scapegoat tree:

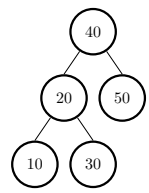

Figure 1: A Scapegoat Tree

Insert the keys $31, 32, 33, 34, 35$ in order. Restructure as needed. Explain all relevant computations. You don't need to show every intermediate tree but show enough to be clear.

Solution:
Inserting 31, 32, and 33 present no issues:
31: $d = 3$, $\log_{3/2} n = \log_{3/2} 6 \approx 4.4$, and $3 \not> 4.4$.
32: $d = 4$, $\log_{3/2} n = \log_{3/2} 7 \approx 4.8$ and $4 \not> 4.8$.
33: $d = 5$, $\log_{3/2} n = \log_{3/2} 8 \approx 5.1$ and $5 \not> 5.1$.
Now we have:

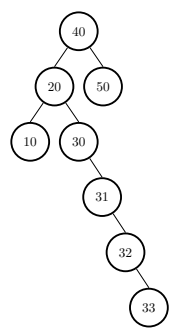

Figure 2: Violation After Inserting 34

Inserting 34 presents an issue:
34: $d = 6$, $\log_{3/2} n = \log_{3/2} 9 \approx 5.4$ and $6 > 5.4$.
Now we have:

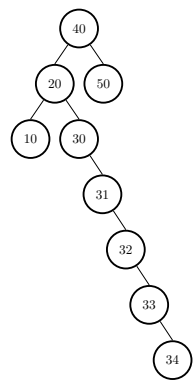

Figure 3: Analysis After Inserting 34

Thus we must restructure. We trace back up the keys to find a scapegoat:
34: $\frac{0}{1} \not> \frac{2}{3}$ so not a scapegoat.
33: $\frac{1}{2} \not> \frac{2}{3}$ so not a scapegoat.
32: $\frac{2}{3} \not> \frac{2}{3}$ so not a scapegoat.
31: $\frac{3}{4} > \frac{2}{3}$ so a scapegoat!
We restructure, now we have:

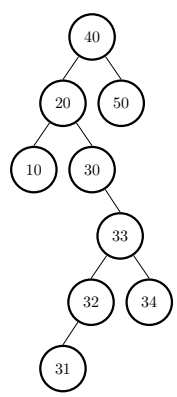

Figure 4: After Inserting 34

Inserting 35 presents no issue:
35: $d = 5$, $\log_{3/2} n = \log_{3/2} 10 \approx 5.7$ and $5 \not> 5.7$.
Now we have:

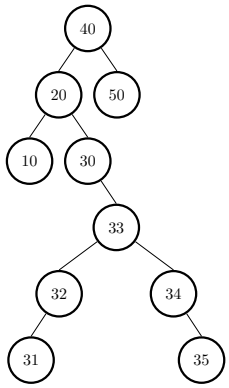

Figure 5: After Inserting 35

2. Regarding scapegoat trees:

   (a) Suppose n is a positive integer. Prove that if n is not a power of 2 then $\lfloor \lg(n-1) \rfloor = \lfloor \lg n \rfloor$.

   Solution:
   To verify this observe that we have $2^k < n < 2^{k+1}$ for some $k \in \mathcal{Z}^+$ and then $k < \lg n < k+1$ and so $k \leq \lfloor \lg n \rfloor < k+1$ and observe that we have $2^k \leq n-1 < 2^{k+1}$ and then $k < \lg(n-1) < k+1$ and so $k \leq \lfloor \lg(n-1) \rfloor < k+1$. Since k is an integer we must have $\lfloor \lg(n) \rfloor = \lfloor \lg(n-1) \rfloor$.

   (b) Use strong induction to prove the theorem from class that states that a restructured tree with n nodes will have height $\lfloor \lg n \rfloor$. Part (a) will probably be useful.

   Solution:
   Clearly the result is true for a tree with 1 node since the height is $0 = \lfloor \lg 1 \rfloor$.
   Assume it is true for all trees with fewer than n nodes. For a tree with n nodes the left subtree will have $\lfloor n/2 \rfloor$ nodes while the right subtree will have $n - 1 - \lfloor n/2 \rfloor$ nodes.
   Since the left subtree has at least as many nodes as the right subtree it will be at least as high, so let's look at that height.
   If n is even then the left subtree has $n/2$ nodes and hence has height:

$$\lfloor \lg n/2 \rfloor = \lfloor \lg n - \lg 2 \rfloor = \lfloor \lg n - 1 \rfloor = \lfloor \lg n \rfloor - 1$$

   If n is odd then the left subtree has $(n-1)/2$ nodes and hence has height:

$$\lfloor \lg((n-1)/2) \rfloor = \lfloor \lg(n-1) - \lg 2 \rfloor = \lfloor \lg(n-1) - 1 \rfloor = \lfloor \lg(n-1) \rfloor - 1$$

   However when n is odd we have $\lfloor \lg(n-1) \rfloor = \lfloor \lg n \rfloor$ so this equals $\lfloor \lg n \rfloor - 1$.
   This means that the height of the entire rebuilt tree is:

$$1 + \lfloor \lg n \rfloor - 1 = \lfloor \lg n \rfloor$$

3. Suppose that at some instant we know that the height of a scapegoat tree satisfies the following with $n \geq 2$:

$$h \leq \left\lfloor \log_{3/2} n \right\rfloor + 1$$

(a) Suppose we insert a node and do not need to rebuild the tree. Thus we now have $n+1$ nodes. Denote the new height by h'.

Prove that the height condition is still valid. That is, prove:

$$h' \leq \left\lfloor \log_{3/2}(n+1) \right\rfloor + 1$$

Solution:
The fact that we do not need to rebuild the tree means that the depth of the inserted node satisfies:

$$d \leq \left\lfloor \log_{3/2}(n+1) \right\rfloor$$

In addition we know (this takes a bit of work, omitted here) that:

$$\left\lfloor \log_{3/2}(n+1) \right\rfloor \leq \left\lfloor \log_{3/2} n \right\rfloor + 1$$

However the original tree had:

$$h \leq \left\lfloor \log_{3/2} n \right\rfloor + 1$$

And thus the new node does not increase the height of the tree beyond this expression and so:

$$h' \leq \left\lfloor \log_{3/2} n \right\rfloor + 1 \leq \left\lfloor \log_{3/2}(n+1) \right\rfloor + 1$$

(b) Suppose we delete a node and do need to rebuild the tree. Thus we now have $n-1$ nodes. Denote the new height by h'.

Prove that the height condition is still valid. That is, prove:

$$h' \leq \left\lfloor \log_{3/2}(n-1) \right\rfloor + 1$$

Solution:
Well observe that by our height analysis for a rebuilt tree:

$$h' = \lfloor \lg(n-1) \rfloor < \left\lfloor \log_{3/2}(n-1) \right\rfloor < \left\lfloor \log_{3/2}(n-1) \right\rfloor + 1$$

4. Suppose we insert keys $1, 2, 3, \ldots$ in that order into an initially empty generalized scapegoat tree with parameter α. Suppose we fix a key k and insist that α be chosen so that k will be the first inserted key which triggers a scapegoat search.

   (a) Find an inequality for α in terms of k.

   Solution:

   When we insert key k we will have depth $d = k - 1$ and hence a scapegoat search will be triggered when:

$$\begin{aligned} k - 1 &> \log_{1/\alpha} k \\ (1/\alpha)^{k-1} &> k \\ \alpha^{k-1} &< \frac{1}{k} \\ \alpha &< \sqrt[k-1]{\frac{1}{k}} \end{aligned}$$

   (b) Take your inequality and make it an equality and then plot this function for $1 \leq k \leq 100$ and explain why its shape makes sense. You should comment both on the general shape (increasing? decreasing?) nature of the graph as well as its behavior as $k \to \infty$.

   Solution:

   Here is the plot:

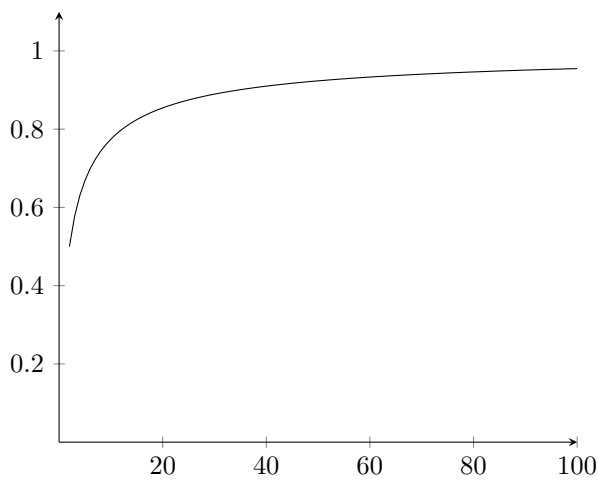

Figure 6: α as a Function of k

   The shape makes sense because as k gets larger we must permit the tree to be more unbalanced before a scapegoat search is triggered. This means α must get larger, closer and asymptotic to 1.

5. Regarding Scapegoat trees, we proved a lemma in class that stated:

   If a node p has $\text{depth}(p) > \log_{3/2} n$ then p has an ancestor u for which:

$$\frac{\text{size}(u.child)}{\text{size}(u)} > \frac{2}{3}$$

   Suppose we replace $\frac{3}{2}$ by $\beta > 1$ and we want generalize the above to:

   If a node p has $\text{depth}(p) > \log_{\beta} n$ then p has an ancestor u for which:

$$\frac{\text{size}(u.child)}{\text{size}(u)} > f(\beta)$$

   (a) Following the proof from class what would be the obvious choice for $f(\beta)$? Explain.

   Solution:

   The proof from class has $\beta = \frac{3}{2}$ yielding $f(\beta) = \frac{2}{3}$ so it seems reasonable that in general we ought to get $f(\beta) = 1/\beta$.

   (b) Prove that the statement is true with this $f(\beta)$.

   Solution:

   Suppose that no node from p to the root is a scapegoat candidate. This means that for every ancestor u from p to the root we have:

$$\frac{\text{size}(u.child)}{\text{size}(u)} \leq 1/\beta$$

   The root has size n and so if we follow the search path the root's child has size at most $(1/\beta)n$, that node's child has size at most $(1/\beta^2)n$, and so on, and so a node at depth i will have size at most $(1/\beta^i)^n$. If d is the depth of p then we know:

$$1 \leq \text{size}(p) \leq \left(\frac{1}{\beta^d}\right) n$$

   Ignoring the middle and solving for d:

$$\begin{aligned} 1 &\leq \left(\frac{1}{\beta^d}\right) n \\ \beta^d &\leq n \\ d &\leq \log_{\beta} n \\ d &\leq \log_{\beta} n \\ \text{depth}(p) &\leq \log_{\beta} n \end{aligned}$$

   This is a contradiction.

6. Suppose we want to reduce the frequency of rebuilding a scapegoat tree on insertion. To do this, should we increase or decrease the value of α? Explain.

   Solution:
   Since we rebuild when the depth is greater than $\log_{1/\alpha} m$ in order to rebuild more frequently we should increase α since this decreases the base which increases the logarithm and makes the scapegoat search happen less frequently.

7. Insert the keys $3, 2, 1, 4, 5, 6, 7, ...$ in that order into an empty scapegoat tree until the first insertion violates the scapegoat height condition. Identify the scapegoat and rebuild. You only need to show:

   - The tree when the violation occurs.
   - Justification of the violation.
   - Justification of the scapegoat.
   - The tree after the rebuild.

   Solution:
   The violation occurs when we insert 9 since then $d = 6 > 5.42 \approx \log_{3/2} 9 = \log_{3/2} n$:

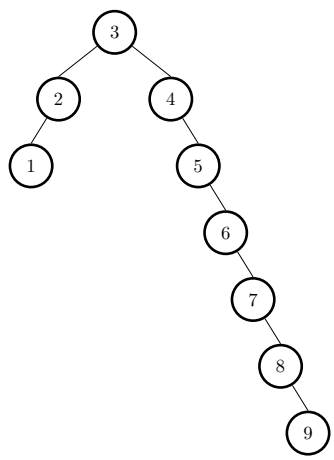

Figure 7: Scapegoat Tree Violation

   We find the scapegoat is the key 6 since $\text{size}(6.right)/\text{size}(6) = 3/4 > 2/3$ and rebuild there:

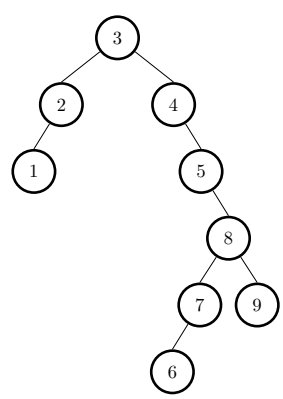

Figure 8: Rebuild at Scapegoat

8. Prove that for a node u in a scapegoat tree, if u violates the scapegoat condition with $\alpha = 2/3$ and with its left child then we must have:

$$\frac{1}{2}\text{size}(u.left) > 1 + \text{size}(u.right)$$

Solution:
Since we have:

$$\frac{\text{size}(u.left)}{\text{size}(u)} > \frac{2}{3}$$

It follows that:

$$\begin{aligned}
\text{size}(u.left) &> \frac{2}{3}\text{size}(u) \\
\text{size}(u.left) &> \frac{2}{3}(1 + \text{size}(u.left) + \text{size}(u.right)) \\
\frac{1}{3}\,\text{size}(u.left) &> \frac{2}{3} + \frac{2}{3}\,\text{size}(u.right) \\
\frac{1}{2}\,\text{size}(u.left) &> 1 + \text{size}(u.right)
\end{aligned}$$

9. This problem works through a specific example of the token calculations associated with the amortized analysis of a Scapegoat Tree. Starting with an empty Scapegoat Tree, Suppose the following sequence of operations is performed where I means insert and D means delete:

$$\{ \text{I:10, I:3, I:1, I:4, I:20, I:30, I:40, I:50, I:60, D:60, I:60, I:70} \}$$

The final insert will trigger a scapegoat hunt and rebuild but for now don't worry about that.

(a) Draw a table showing the progress of the ID account balance during the sequence of operations.

Solution:
Observe that we are performing $k = 12$ operations and so each insert and delete gives $\lfloor \log_{3/2} 12 \rfloor + 3 = 9$ tokens to the ID account and each insert and delete takes from that account an amount equal to $1 + d$ where d is the depth of the inserted or final deleted node.
Here is the ID account statement:

Operation	Deposit	Depth	Cost	Balance
I:10	9	0	1	8
I:3	9	1	2	15
I:1	9	2	3	21
I:4	9	2	3	27
I:20	9	1	2	34
I:30	9	2	3	40
I:40	9	3	4	45
I:50	9	4	5	49
I:60	9	5	6	52
D:60	9	5	6	55
I:60	9	5	6	58
I:70	9	6	7	60

(b) How many tokens are in the DR account at the end? Explain.

Solution:
Since we only deleted one node the DR account has $1(3) = 3$ tokens.

(c) Draw a picture of the tree and next to each node show the totals for the IR account at the end.

Solution:

We have:

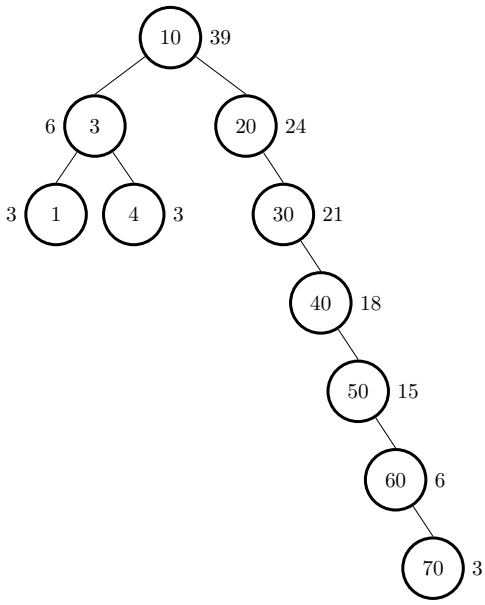

Figure 9: IR Account Illustration

(d) Find the scapegoat and explain how there will be enough tokens to cover the rebuild.

Solution:

The scapegoat will be the node with the key 40 in it. Since the subtree rooted at this node contains 4 keys it will cost 4 tokens to rebuild it and the node's account has 12 tokens which is sufficient.

10. Suppose we insert the keys $16, 97, 81, 63, 35, 72, 55, 41, 2, 34$ into a empty scapegoat tree in that order. We then start deleting keys in the same order until the first complete restructure occurs. We then restructure and stop. What will the tree look like after the restructure?

    Solution:
    At the end of this insertion we have $n = m = 10$. We delete until $n < (2/3)m$ which is when $n = 6$. This means we deleted 4 keys leaving us with $35, 72, 55, 41, 2, 34$. In order these are $2, 34, 35, 41, 55, 72$ and so the restructured tree looks like:

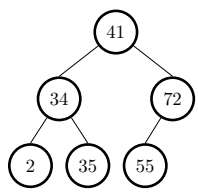

Figure 10: Restructure Due to Deletion

11. Suppose we have a scapegoat tree with unknown α. Suppose we insert the keys $0, 1, 2, 3, ...,$ in that order and key $k = 10$ causes a depth violation. Find the tightest possible upper bound on α.

For $k = 10$ we have $\alpha <$	$\frac{1}{\sqrt[10]{11}}$

Solution:
We know that for $k = 10$ that key is at depth 10 and there are 11 keys and hence we have:

$$\begin{aligned} 10 &> \log_{1/\alpha}(11) \\ \log_{1/\alpha}(11) &> 10 \\ (1/\alpha)^{10} &> 11 \\ (1/\alpha) &> \sqrt[10]{11} \\ \alpha &< \frac{1}{\sqrt[10]{11}} \end{aligned}$$

12. In the following scapegoat tree the key 60 has just been inserted and it violates the height condition. Identify the scapegoat which will be used to fix the issue (justify), and show the rebuild.

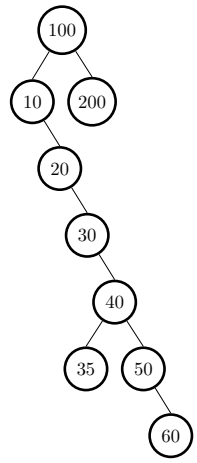

Figure 11: Scapegoat Tree Violation

Solution:
Note that $m = 10$ and $d = 6 > \log_{3/2} 10 \approx 5.68$ which justifies restructuring.
Check the size ratios:

- 50: $\frac{1}{2} \not> \frac{2}{3}$
- 40: $\frac{1}{2} \not> \frac{2}{3}$
- 30: $\frac{4}{5} > \frac{2}{3}$ so restructure at 30.

The result is:

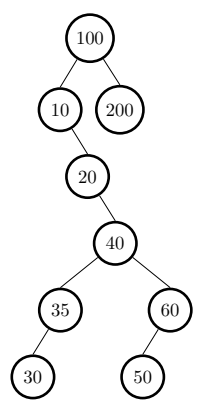

Figure 12: After Rebuild

13. Suppose we insert a key into a scapegoat tree with $\alpha = 2/3$ and it causes a depth violation. Is it possible for there to already be a key in the tree at the same depth? Explain.

Solution:
Yes, it is possible. This will happen if we insert $2, 3, 4, 5, 1, 0, 7, 8$ in that order, then delete $0, 1$ in that order and then insert 6. It's a good idea to work out the details.

14. Suppose we insert a key into a scapegoat tree with $\alpha = 2/3$ and it causes a depth violation. Is it possible for there to already be a key in the tree which is deeper? Explain.

Solution (Originally by Anna Evtushenko):
No, it is not possible. Suppose a scapegoat tree contains n nodes. Due to the height bound on the tree if d_e is the deepest possible node in the tree then we know that:

$$d_e \leq \left\lfloor \log_{3/2} n \right\rfloor + 1$$

Now suppose we insert a new key which causes a depth violation. If d_i is the depth of the inserted node then since there are now $n+1$ keys in the tree this means that $d_i > \log_{3/2}(n+1)$.
Suppose by way of contradiction that $d_e \geq d_i + 1$. Then we have:

$$\left\lfloor \log_{3/2} n \right\rfloor + 1 \geq d_e \geq d_i + 1 > \log_{3/2}(n+1) + 1$$

However since $\log_{3/2} n \geq \left\lfloor \log_{3/2} n \right\rfloor$ this means that:

$$\begin{aligned} \log_{3/2} n + 1 &> \log_{3/2}(n+1) + 1 \\ \log_{3/2} n &> \log_{3/2}(n+1) \\ n &> n+1 \end{aligned}$$

This is a contradiction.

15. Suppose we have a scapegoat tree with parameter $\frac{1}{2} < \alpha < 1$.

(a) As α increases does the frequency of rebuilds due to inserts increase or decrease?

Write the word INCREASE or DECREASE below.

Which?	DECREASE

(b) As α increases does the frequency of rebuilds due to deletions increase or decrease?

Write the word INCREASE or DECREASE below.

Which?	INCREASE

16. Suppose we have a scapegoat tree. We insert m keys and then delete keys until the instant when we have $n < \alpha \cdot m$ keys and a rebuild is triggered. For each of the following α and m, provide n. [5 pts]

α	m	n
$\frac{2}{3}$	9	5
$\frac{4}{5}$	9	7

17. Suppose we have a scapegoat tree with $\alpha = \frac{2}{3}$. We insert m keys and then delete keys until the instant when we have 9 keys and a rebuild is triggered. What could m have been? If you have more than one answer please list them in increasing order.

Possible m	$14, 15$

Scratch Work:
We need $9 < (2/3)m$ so $m > 13.5$ so $m \geq 14$. This is necessary but not sufficient, however, since n needs to be the largest.
Clearly $m = 14$ works since a rebuild occurs when $n < (2/3)14 = 28/3$ so $n = 9$ will cause a rebuild.
Clearly $m = 15$ works since a rebuild occurs when $n < (2/3)15 = 10$ so $n = 9$ will cause a rebuild.
However $m = 16$ does not work since a rebuild occurs when $n < (2/3)16 = 32/3$ so $n = 10$ will cause a rebuild. Similarly other $m > 16$ will cause rebuilds for $n > 9$.

18. Suppose we have a scapegoat tree of height h with unknown $\frac{1}{2} < \alpha < 1$ which is perfect, having $n = 2^{h+1} - 1$ keys. Prove that inserting a key into this scapegoat tree could not possibly trigger a scapegoat-hunt.

Solution:
When we insert a key it will be at depth $h+1$ and we will have $n = 2^{h+1}$. This means that in order to trigger a scapegoat-hunt we would need:

$$\begin{aligned} h+1 &> \log_{1/\alpha} 2^{h+1} \\ (1/\alpha)^{h+1} &> 2^{h+1} \\ 1/\alpha &> 2 \\ \alpha &< \frac{1}{2} \end{aligned}$$

This is a contradiction.

Note that there could be other approaches to this which I haven't thought of.

19. Consider the following scapegoat tree:

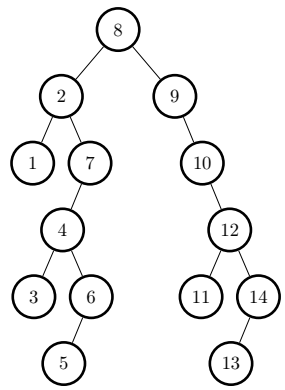

Figure 13: A Scapegoat Tree

(a) Identify the nodes which are scapegoats with $\alpha = 2/3$.
Solution:
The scapegoats are $1, 2, 9, 10$.

(b) For which α would there be no scapegoats at all?
Solution:
The largest associated ratio (at node 9) is $5/6$ so provided $\alpha \geq 5/6$ there will be no scapegoats.

20. Suppose a binary tree contains these keys:

$$4, 5, 10, 12, 13, 20, 21, 30, 50, 55$$

If we rebuild this tree, what will it look like?

Solution:
Here is the rebuilt tree:

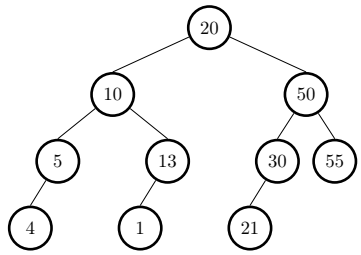

Figure 14: After Rebuild

21. Suppose we insert and delete nodes into a scapegoat tree according to the following pattern:

I,I,I,I,I,D,D,I,I,D,D,D,I,I,I,I,I,I,D

For this problem, ignore all rebuilding! After each operation what are the values of n and m?

Solution:
We have:

Operation	n	m
I	1	1
I	2	2
I	3	3
I	4	4
I	5	5
D	4	5
D	3	5
I	4	5
I	5	5
D	4	5
D	3	5
D	2	5
I	3	5
I	4	5
I	5	5
I	6	6
I	7	7
I	8	8
D	7	8